

Auto-Parallel Technologies Externship Report

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Introduction & Overview

This externship with Auto-Parallel Technologies in Austin was a unique blend of software engineering, open source exploration, and hands-on tech immersion. My goal going in was simple: to understand how real-world software is built, released, and used, and how different elements like hardware, AI models, and open source systems connect in the industry.

Beyond just programming, I hoped to experience the culture behind it all, how developers think, how teams collaborate, and how technologies evolve. And that's exactly what I got, along with more than a few surprises.

Day 0, Sunday: Arrival & Preparation

The trip started chaotically. We were delayed for our flight, and after a sleepless night, I landed in Austin with a splitting headache around 10:30 a.m. Since our Airbnb wouldn't let us in until 4 p.m., we searched for a place to store our luggage and grab food. After some wandering, we ended up at the LBJ Presidential Library.

With our bags checked in (thanks to buying tickets), we explored the museum. We saw relics from the Vietnam War, soldier letters, old currency, a presidential limousine, the original Oval Office desk, and even interacted with a talking LBJ puppet (which sadly refused to shake hands). They had modulated call stations where I technically had a conversation with the IBM president, a surreal moment. By the time we got to the Airbnb, I was completely knocked out.

Day 1, Monday: PerlGPT and Diogenes Hackerspace

We started the day at Mr. Braswell's house. He gave us a pep talk about Scout Law, basic expectations, and the significance of open source software. We went over Linux concepts, pronunciation nuances, and LAMP architecture basics.

Later, we visited the InMotion Software team at their WeWork office. There, we met Mr. Jason, Mr. Greg, Ms. Kimberly, Mr. Nathan, and even the CEO, Mr. John. Mr. Jason taught us about the Iron Triangle, balancing scope, cost, and time, a constant challenge in software management. All incredibly generous with their time and insights.

Mr. Braswell had us try some programming exercises and, well, I accidentally hosed my Linux OS. One command and everything broke. It was unfixable, so I had to reinstall the OS from scratch. That was definitely a milestone.

We also learned the full Software Development Life Cycle (SDLC), from purpose and planning to deployment and maintenance. Mr. Braswell introduced us to Perl's history and how modern languages like PHP and Ruby drew heavily from it.

Mr. Braswell also showed us his Perl-based LLM project and explained how reverse engineering old Perl modules on CPAN worked. We saw how good variable naming and documentation can help decode unfamiliar code without comments. We also watched a live client demo with Mr. Braswell and Mr. John, showing how communication across tech and non-tech roles happens in practice.

Later that day, we visited Diogenes Hackerspace, an awesome maker space with FPGA chips, 3D printers, laser cutters, and tools I'd never seen back home. Mr. Rob gave us a quick tour. It was the kind of place where ideas turn real.

Day 2, Tuesday: ShinyCMS, Re:3D, and Austin Downtown

We began at Mr. Braswell's place again and then headed to the Hackerspace. That day, we looked into his project ShinyCMS, a Perl-based content management system similar to WordPress. He walked us through live testing, deployment prep, and managing dependencies.

In the afternoon, we toured Re:3D Inc., led by Mr. Michael Pregill. We saw industrial 3D printers like the Gigabot XLT and CNC machines. Learned how extrusion works, how layers build up designs, and how dangerous CNC machines can be if mishandled. The coolest part? They're working with NASA on printing in space using waste materials, super advanced stuff we weren't allowed to photograph.

After a Chipotle lunch, we toured downtown Austin, the historic Driskill Hotel, 6th Street with its music bars and shops, and of course cowboy hats courtesy of Mr. Braswell. We saw the spot where his friend performs magic tricks, you can even catch the show from outside the window without needing a ticket. The downtown was beautiful and walkable, full of towering skyscrapers and vibrant scenes. While heading to visit the Capitol at night, I even had a fun conversation with two British fellows I ran into, a really cool, spontaneous interaction.

We ended the night at Congress Bridge to catch the bats and view the skyline, followed by a quick peek at the Capitol building before heading home.

Technically, we covered a lot: Perl hashes, how they translate to unordered maps in C++, hash functions, RNGs, memory allocation, vectors, and a general discussion about how open source has evolved and sometimes stagnated due to gatekeeping.

Day 3, Wednesday: Navi AI, Docker, and Perl on Xubuntu

We kicked off the final day with another visit to Mr. Braswell. Then headed to Diogenes Hackerspace, where he finally showed us his project, Navi OS. It runs entirely on his own LLM which includes a physical robot.

The OS is built on Docker Compose, running three containers, one for Perl, one for Llama 13b (core LLM), and one for TensorFlow-based image classification. Together, they formed a unified AI operating system running on Ubuntu. We got to observe how image recognition is implemented and how it interfaces with the OS, which used Apache to provide a local web interface

We also had some fun chatting with Navi. It was quirky, surprising, and even a little creepy at moments. It was a first-hand look at how LLMs can sometimes hallucinate, producing fictional memories or interactions. We learned about the difference between GPU memory usage and GPU processor utilization when handling large models.

We edited .bashrc system file to auto-load CPAN libraries on boot, and also learned how PostgreSQL can be used as a vector-based DB for parallel data operations. We also saw the first model of Navi, the Navi I100.

There were more Perl exercises like regular expressions, encoding and decoding strings, and converting data types. We ended with a photo session next to the robot. Sadly, we didn't get to see it in action, but next time hopefully.

Debrief & Conclusion

Roses: I loved the hands-on learning, exploring open source in action, and seeing real engineers build robots, AI, CMS platforms, and giant 3D printers. The people, Mr. Braswell, Mr. Rob, Mr. Michael, all shared knowledge like true mentors.

Thorns: The Texas heat was brutal. I also wish I had brushed up more on Docker and Perl before arriving. I could've asked better questions.

Buds: Next time, I want to dive deeper into AI model training and hardware integration. I'm curious about how inference pipelines are optimized and how multiple containers work in harmony.

In the end, yes, I definitely achieved what I hoped for. I left with a clearer view of the software world, deeper technical insight, and a fire to keep building. If you're considering externships like this, go for it. Be curious, ask questions, and enjoy every bit of it.